8	AC Voltage	AC Supply		

Precautions:

1. When measuring DC voltage, care must be taken while connecting the two probes of the meter with the DC source. It should be connected with correct polarities in parallel connection with DC supply. That is, positive probe of the meter should be connected to positive terminal of the DC supply and negative probe to negative terminal of the DC supply.
2. When measuring DC current, multimeter should be connected in series with the DC supply and the load with correct polarities. That is, positive probe should be connected with positive terminal of the supply and negative probe to the load.
3. In the above two measurements, if we wrongly connect the probes, the meter needle will get reversed and subsequently the meter will get damaged.
4. While measuring AC voltages, it should be connected in parallel but polarity need not be considered.
5. While measuring AC current, as like the measurement of DC current, the meter should be connected in series with the supply and the load. Polarity need not be considered as like in measuring DC voltage.
6. We should not measure voltage or current in ohm's range. Otherwise, the meter will become faulty.
7. While measuring, the angle of vision towards the meter should be straight. Otherwise, parallax error will result in the measuring quantities.
8. We should not measure high voltage or current by keeping one probe at the supply and the other probe's metal point in contact with our hand, which will result in fatal electric shock.
9. We should presume the voltage and current to be measured at a point of circuit. If any doubt arises, Keep the meter in high range and then measure. If it is a low voltage, then we can select the lower ranges for measuring.
10. High voltage must not be measured using low voltage range. Otherwise the meter will become faulty.
11. Before measuring Ohms, both probes have to be short circuited and adjust pointer to 0 Ω, in order to get accurate values.
12. Select X 1 range to check continuity of a power cord or any wire.
13. In meters, DC is denoted by ⎓.
14. In digital multimeters, if we measure negative voltage or current, it shows reading with a negative sign. In analog meters, negative values should be measured by reversing the meter probes.

EE_Practicals_English Version.indd 239 11/10/2020 12:17:45 PM